GECO: evidence, intervention and recovery

RESEARCH QUESTION

A reproducible simulation and methods study

Corrected prediction · matched comparisons · a measurable saline rig

IEEE-style research seminar | team and author details to be completed

The motivating question

WHY THIS TOPIC

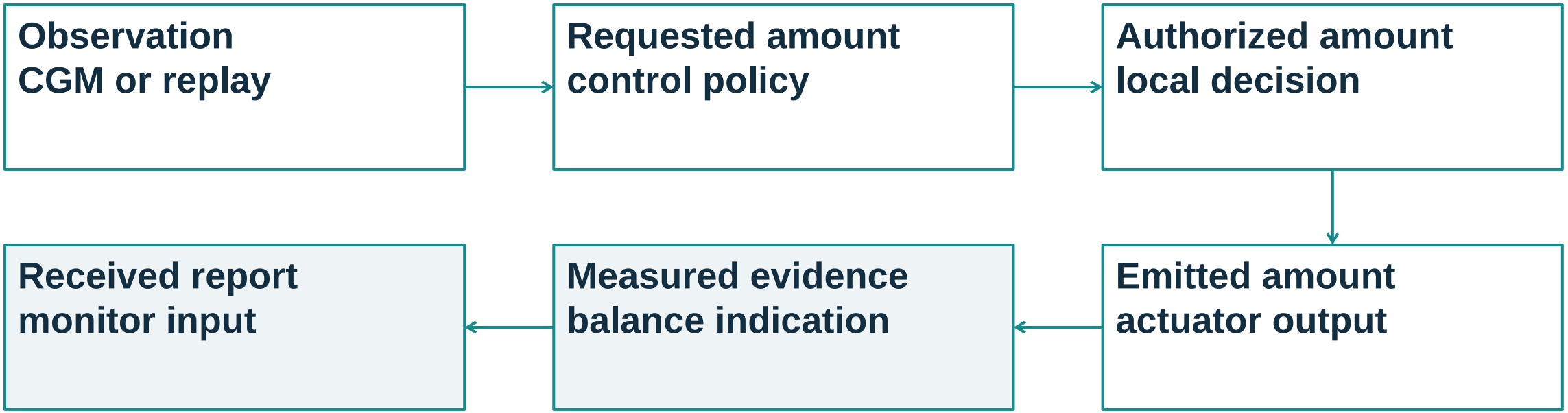
A permitted command and an accepted command do not prove fluid reached the outlet.

A monitor can reduce unwanted delivery while also interrupting wanted delivery.

The study measures both sides of intervention, including return to service.

The evidence boundary

WHY THIS TOPIC



Each link has its own value, timestamp, authority and possible failure.

The contribution to test

CONTRIBUTION

An auditable chain from observations and commands to measurement and response.

A comparison that matches what each practical method can observe and control.

A release that preserves failures, censored recovery and model-domain violations.

What each evidence source can establish

EVIDENCE

Evidence	Supported use	Current limit
Supplied participant records	Internal forecast comparison	Previously exposed; not external validation
New synthetic trajectories	Paired intervention experiments	Depend on plant and actuator assumptions
Native rig animation	Explain proposed components and measurements	Conceptual, not an as-built rig
Future saline measurements	Physical measurement and stopping estimates	IIT Patna measurement pending

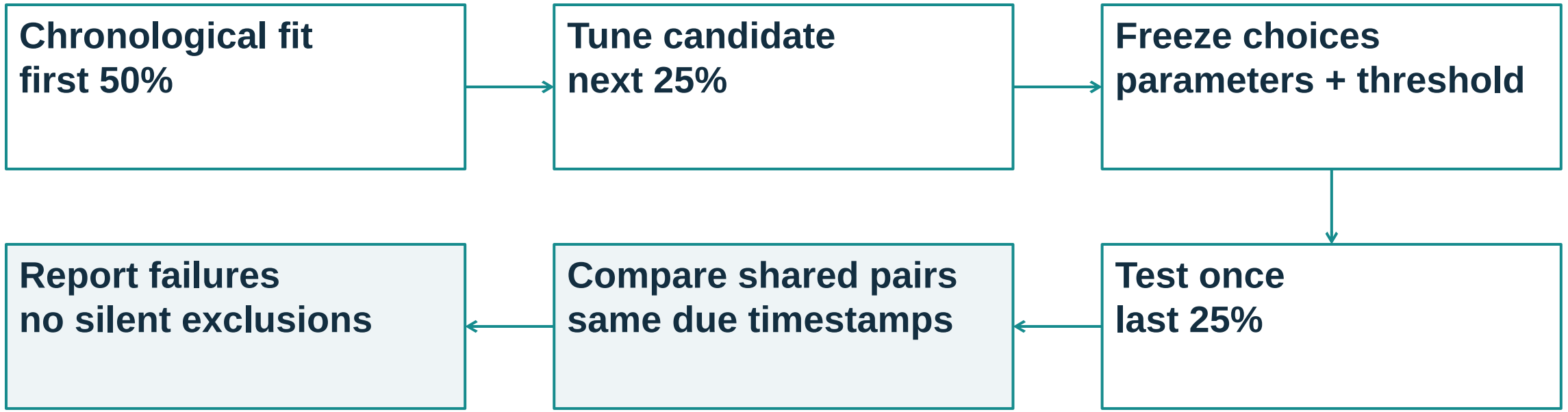
Corrections before new experiments

METHODS

Problem	Repair	Validation
Uncertain covariance and gaps	Joseph update, finite and PSD guards, gap reset	Focused failure and timing tests
Prediction choice	Causal mechanistic, persistence and trend candidates	Fit/tune/test separation
Calibration path	One frozen runtime configuration; explicit failures	Configuration and execution receipts
Response accounting	Previous inputs; interval timing; actual emitted input	Independent trace recomputation

Calibration and causal prediction

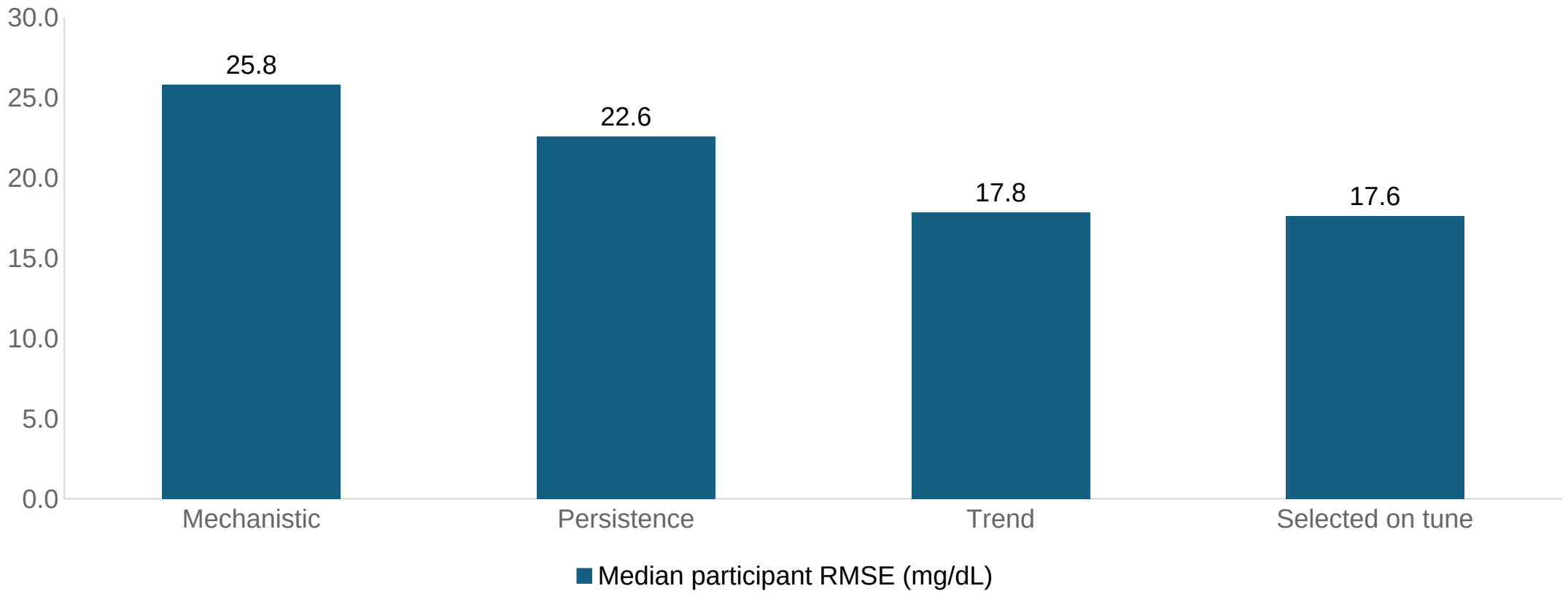
METHODS



Only information available at the prediction origin can enter that forecast.

Thirty-minute prediction error

PREDICTION RESULTS



31 participants; 77,110 shared pairs. Lower is better. Medians weight participants equally.

Prediction findings and limits

PREDICTION RESULTS

Finding	Estimate	Interpretation
Selected − persistence RMSE	−4.14 mg/dL; 95% CI −5.02 to −3.33	Paired participant bootstrap
Mechanistic versus persistence	Worse in 28 of 31 participants	Added structure did not ensure accuracy
Test union alarm burden	13 of 31 exceeded 1 episode/day	Tune budget did not generalize to all
Record disposition	31 evaluated; 2 excluded; 1 failed	Negative insulin entries caused explicit failure

What this says about the model

PREDICTION RESULTS

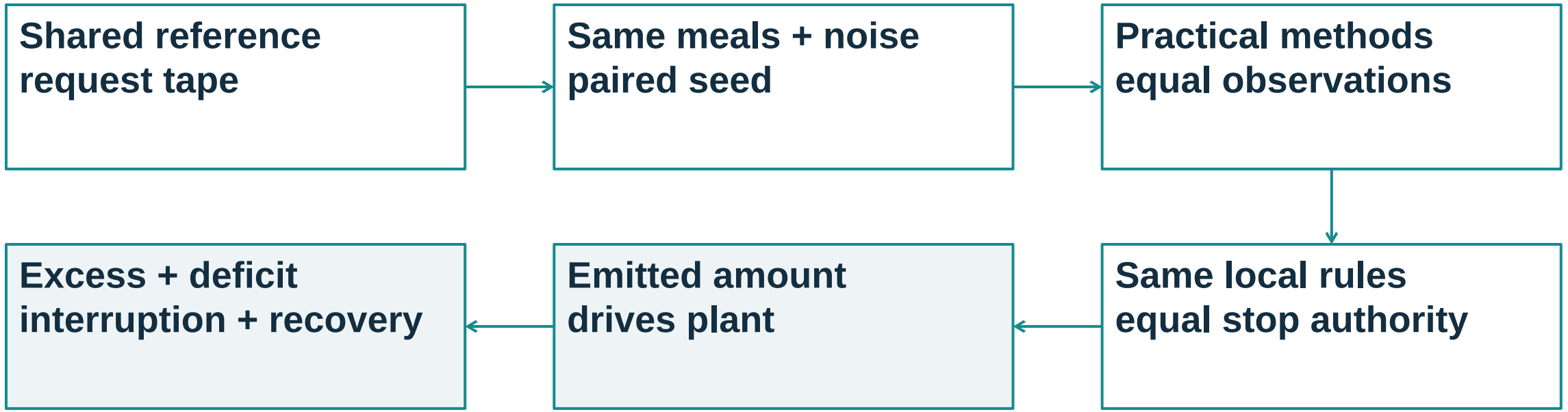
The mechanistic model was less accurate than persistence for most supplied participants.

The corrected pipeline therefore permits a simpler predictor selected before testing.

This is a limit of this model, implementation and dataset combination; it is not a verdict on every physiological model.

Matched intervention experiments

METHODS



Oracle uses evaluator-only fault timing and is reported as a privileged benchmark.

The primary experiment

METHODS

Design choice	Frozen specification
Physiological plants	GECO original five-state; original Hovorka 2004 equations
Test units	24 parameter vectors × 2 noise seeds per plant
Duration and disturbances	48 h at 5 min; fault 12–14 h; eight conditions
Primary comparisons	Six response arms; 4,608 runs
Stop-model sensitivity	576 runs; 5,184 completed in total

Faults and observation assumptions

METHODS

Condition	Injected change	Evidence retained
Excess / stuck-on	Actuator input doubled / added emission	Local and reported balance
Occlusion	No emission in fault window	Commands and local balance
Biased / missing CGM	+40 mg/dL / absent report	Mass evidence and local rules
Combined excess + bias	Excess with +25 mg/dL CGM bias	Local and reported mass
Masked network report	Combined fault; report substituted	Independent local mass remains

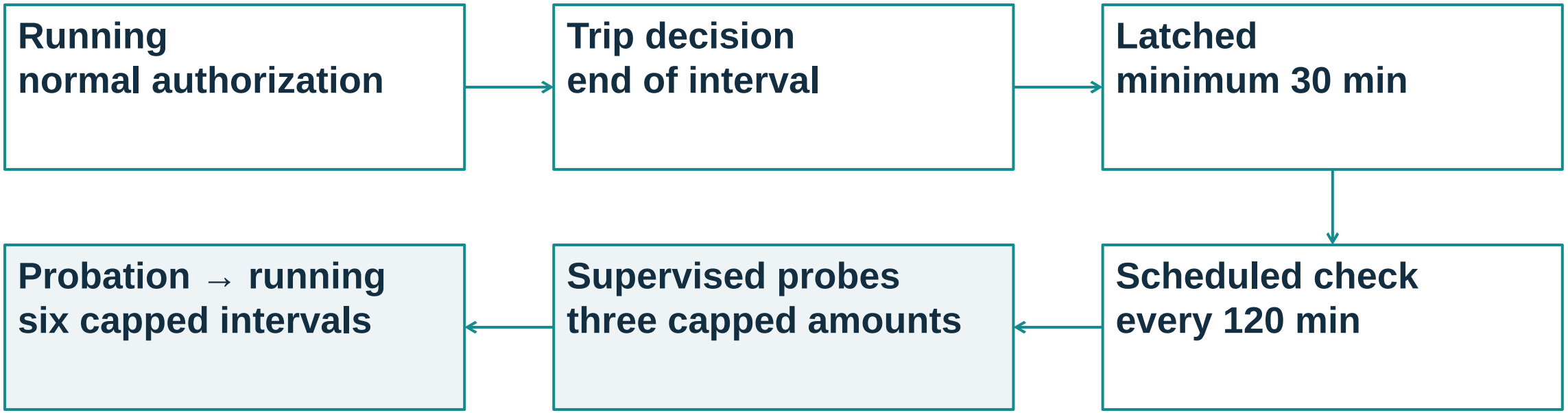
The six primary response arms

METHODS

Arm	Monitor	Return to service
Full, manual	Mass CUSUM + physiological residuals	Scheduled manual release
Full, supervised	Same full detector	Probe and probation
Simple, manual	Instantaneous absolute mass residual	Scheduled manual release
Simple, supervised	Same simple detector	Probe and probation
Sham / oracle	Local rules only / known fault timing	Matched local rules / privileged timing

Response and recovery states

METHODS



No catch-up delivery. Manual release and supervised recovery are separate comparisons.

Outcomes that a stop can hide

METHODS

Outcome	Definition	Reason to retain
Stepwise excess	Sum max(emitted − clean, 0)	Extra delivery cannot cancel a deficit
Stepwise deficit	Sum max(clean − emitted, 0)	A long stop remains visible
Longest interruption	Positive request; zero emission	Duration matters beyond total dose
Recovery onset / confirmation	First six good intervals / confirmation 30 min later	No single ambiguous recovery time
Censored recovery / domain flags	Retain failed outcomes and denominators	Failures remain in the denominator

Primary experiment outcome

PRIMARY RESULTS

5,184 runs completed without execution exceptions.

The full monitor did not show consistent superiority to the simpler monitor.

Independent review checked every full trace and recomputed core outcomes; all domain violations were retained.

Selected paired contrasts

PRIMARY RESULTS

Full supervised − simple supervised	Mean difference	95% paired CI
Original: stuck-on excess	+0.13745 research U	+0.10044 to +0.17445
Hovorka: stuck-on excess	+0.08458 research U	+0.04229 to +0.12688
Hovorka: occlusion deficit	+1.13031 research U	+0.13982 to +2.37804

Positive means more excess or deficit for the full monitor. Hovorka results are validity-limited.

Physiological validity audit

PRIMARY RESULTS

Branch	Clean practical comparison	Interpretation
Original model	0/48 clean trips for full and simple	No glucose-domain failure; action clamps remain
Original Hovorka equations	44/48 clean runs had negative EGP	Clean baseline already outside valid domain
Hovorka clean trips	Full 2/48; simple 0/48	False intervention did not disappear
Main Hovorka traces	2,108/2,304 had negative EGP	Intervention benefit claims are restricted

A separate validity and component check

EXPLORATORY SUPPLEMENT

Exploratory follow-up after the primary model-domain failures.

New fasting parameter vectors and noise seeds; actuator-aware basal compensation.

Full, simple, mass-CUSUM-only and sham arms isolate a narrower component question.

Fasting supplement: validity and false stops

EXPLORATORY SUPPLEMENT

Finding	Locked exploratory result
Execution	2,048/2,048 completed; no failed runs
Physiological checks	0 domain-invalid and 0 negative-EGP runs
Original clean trips	Full, mass-CUSUM and simple:3/32 each
Hovorka clean trips	Full 3/32; mass-CUSUM and simple 2/32
Interpretation	Declared screens passed; false stops remain

Full versus mass CUSUM: benefits and costs

EXPLORATORY SUPPLEMENT

Full − mass-CUSUM only	Mean difference	95% paired CI
Original: combined excess+bias excess	−0.07073 research U	−0.09821 to −0.04298
Hovorka: combined excess+bias excess	−0.04187 research U	−0.07946 to −0.00952
Original: bias-only deficit	+1.33868 research U	+0.71855 to +2.10441
Hovorka: bias-only deficit	+0.39818 research U	+0.20601 to +0.60903

Exploratory fasting comparison. Less excess under a combined fault; more deficit under biased CGM.

Reading the simulation movies

RESULT PLAYBACK

Each frame advances through the saved five-minute samples; no synthetic intermediate measurements.

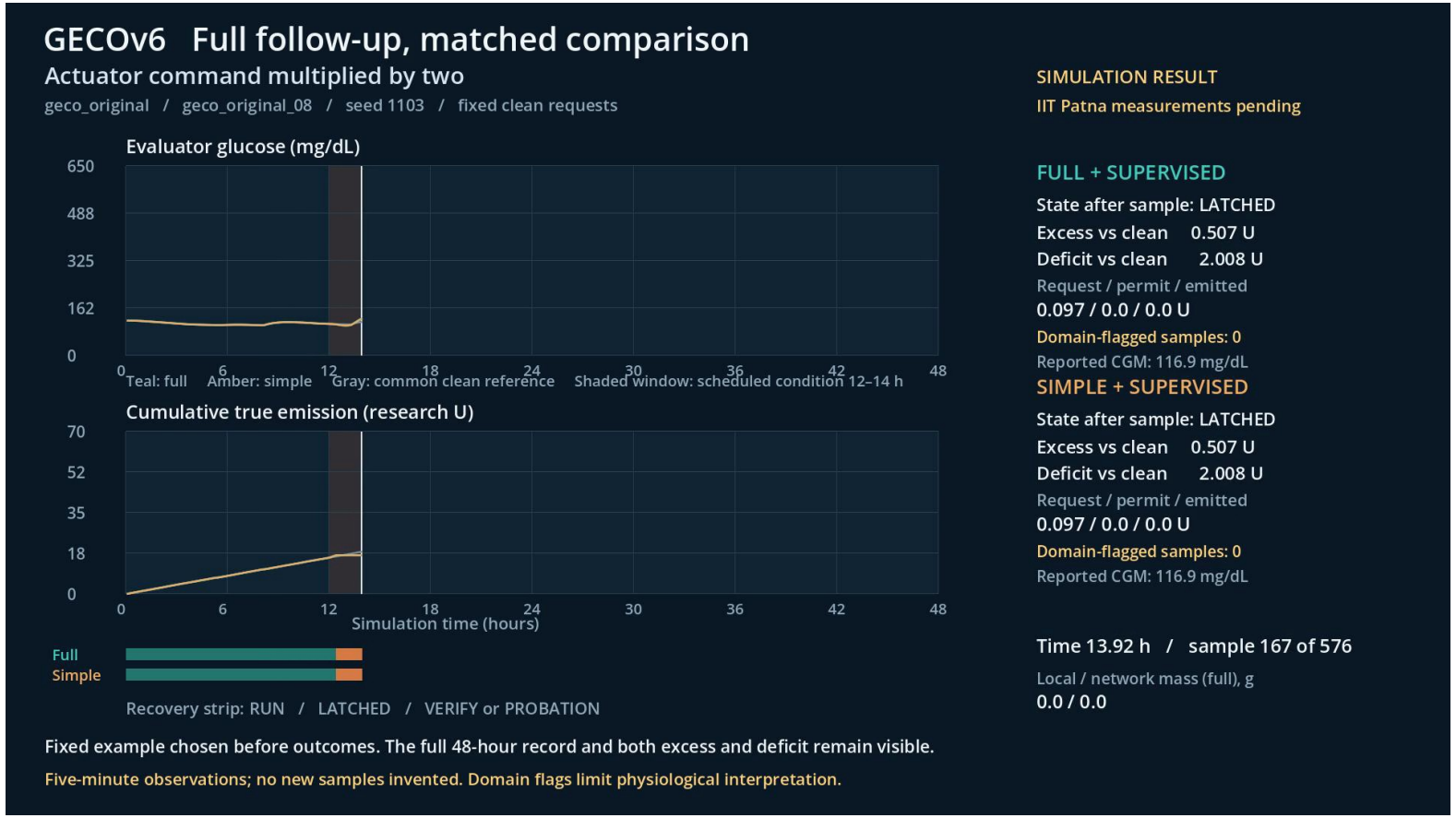
Full and simple monitors share the request tape, noise and reference.

State, emission, excess, deficit and model-domain flags remain visible.

Doubled actuator input: original model

RESULT PLAYBACK

VIDEO IN POWERPOINT

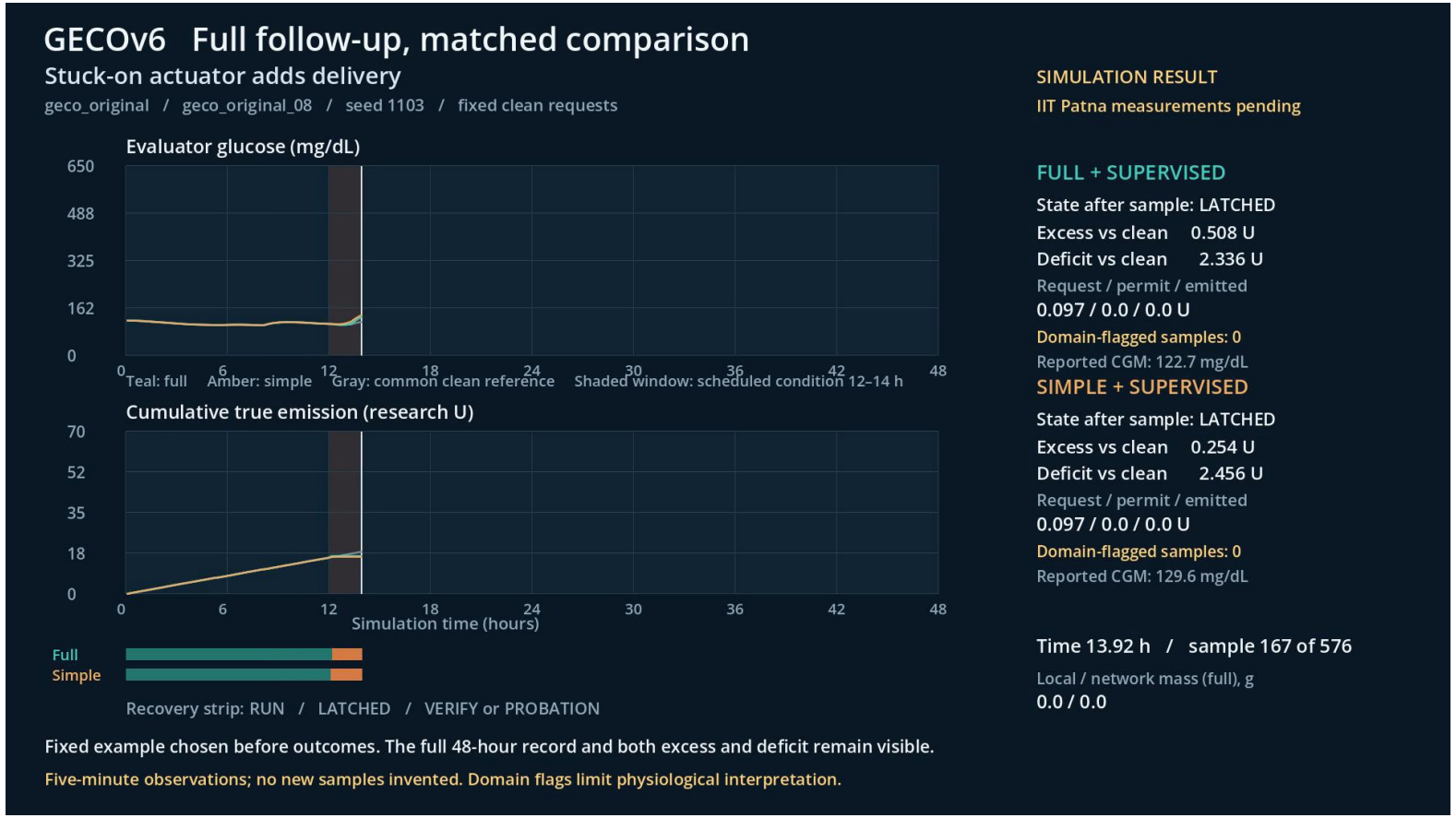


Static poster; video in PowerPoint. Numerical playback only; all displayed quantities come from the frozen trace.

Stuck-on delivery and recovery

RESULT PLAYBACK

VIDEO IN POWERPOINT

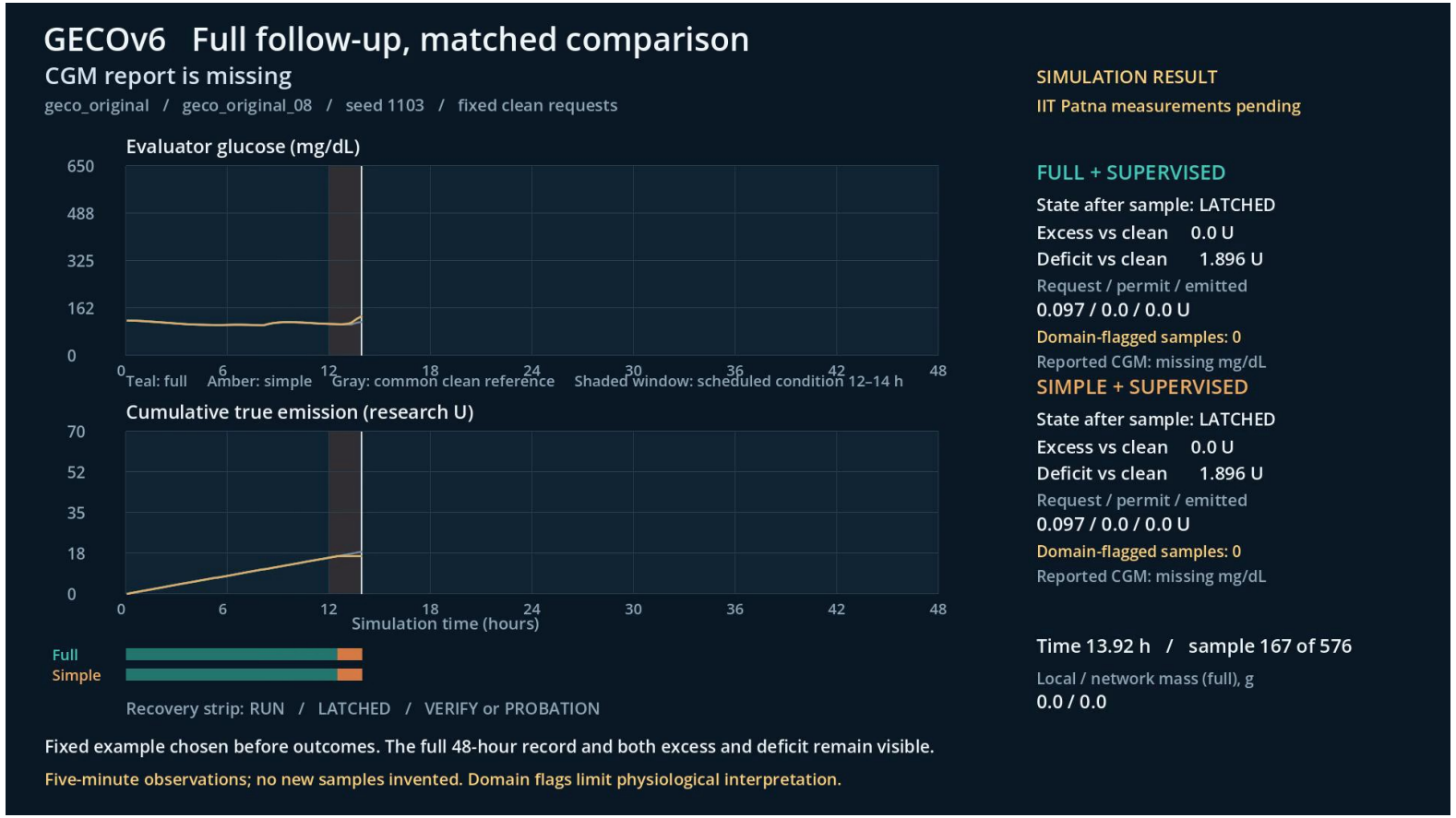


Static poster; video in PowerPoint. Numerical playback only; all displayed quantities come from the frozen trace.

Missing observation and interruption

RESULT PLAYBACK

VIDEO IN POWERPOINT

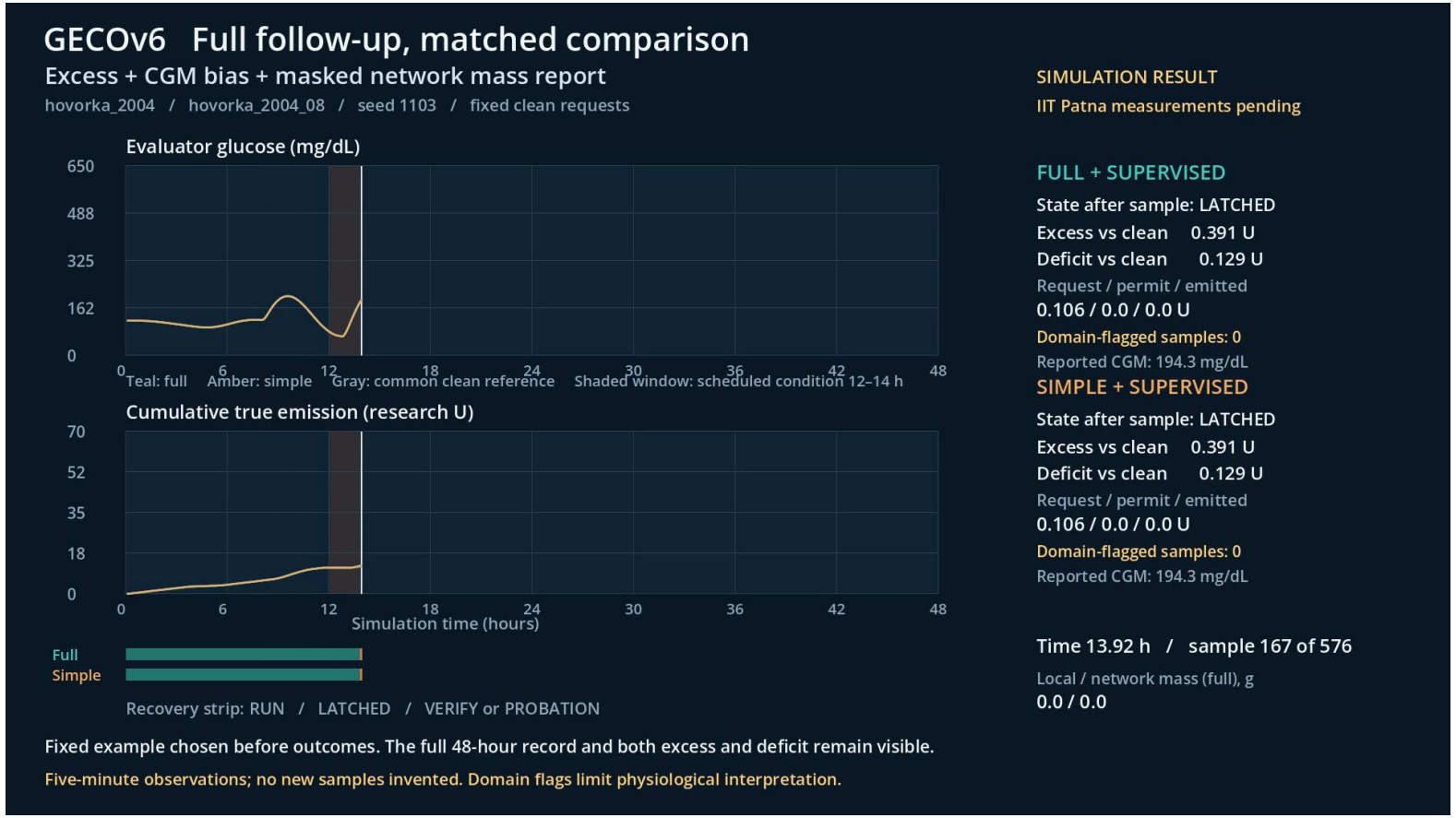


Static poster; video in PowerPoint. Numerical playback only; all displayed quantities come from the frozen trace.

Combined fault: Hovorka domain flags

RESULT PLAYBACK

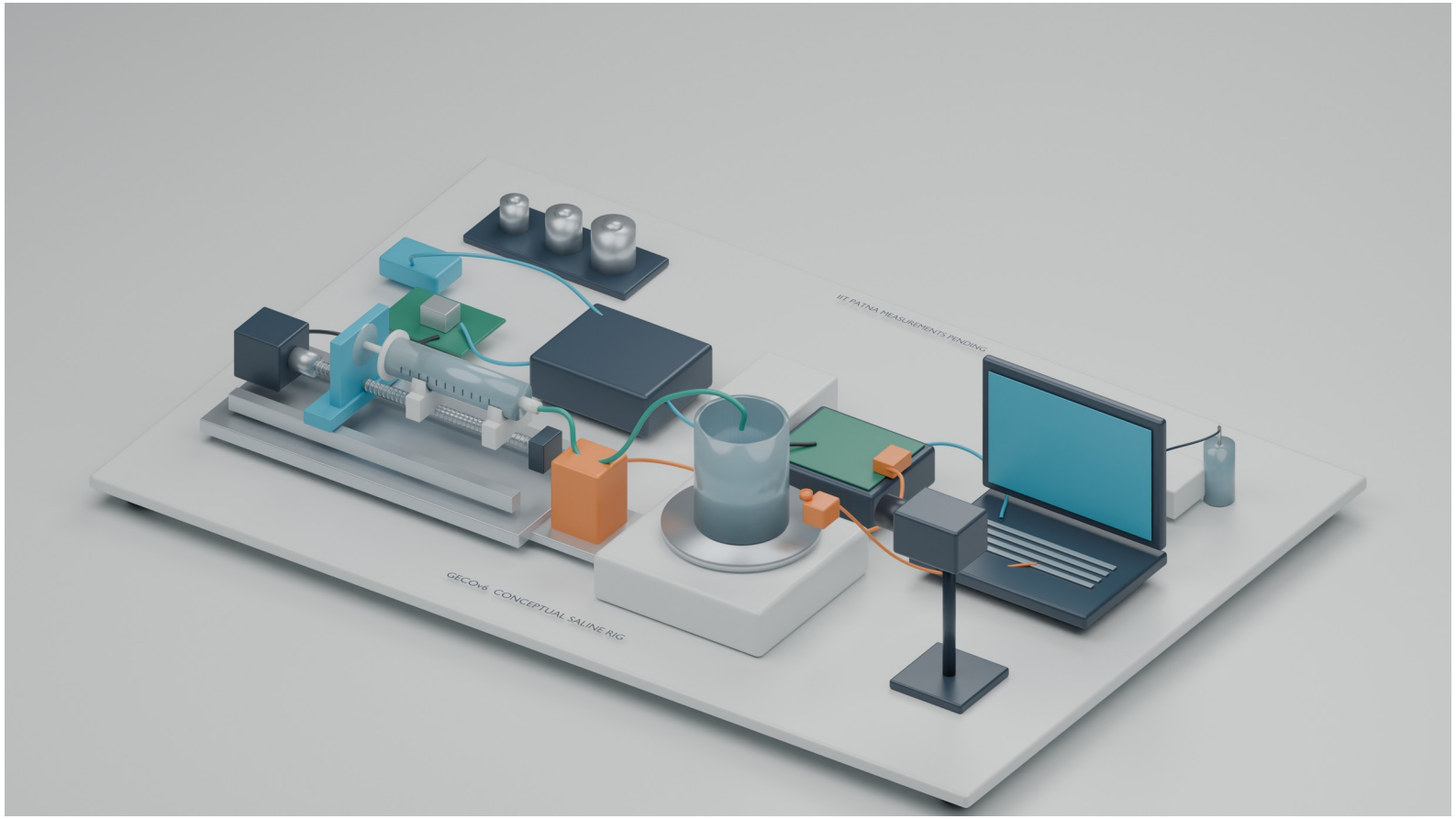
VIDEO IN POWERPOINT



Static poster; video in PowerPoint. Numerical playback only; all displayed quantities come from the frozen trace.

The proposed saline twin

RIG AND MEASUREMENT



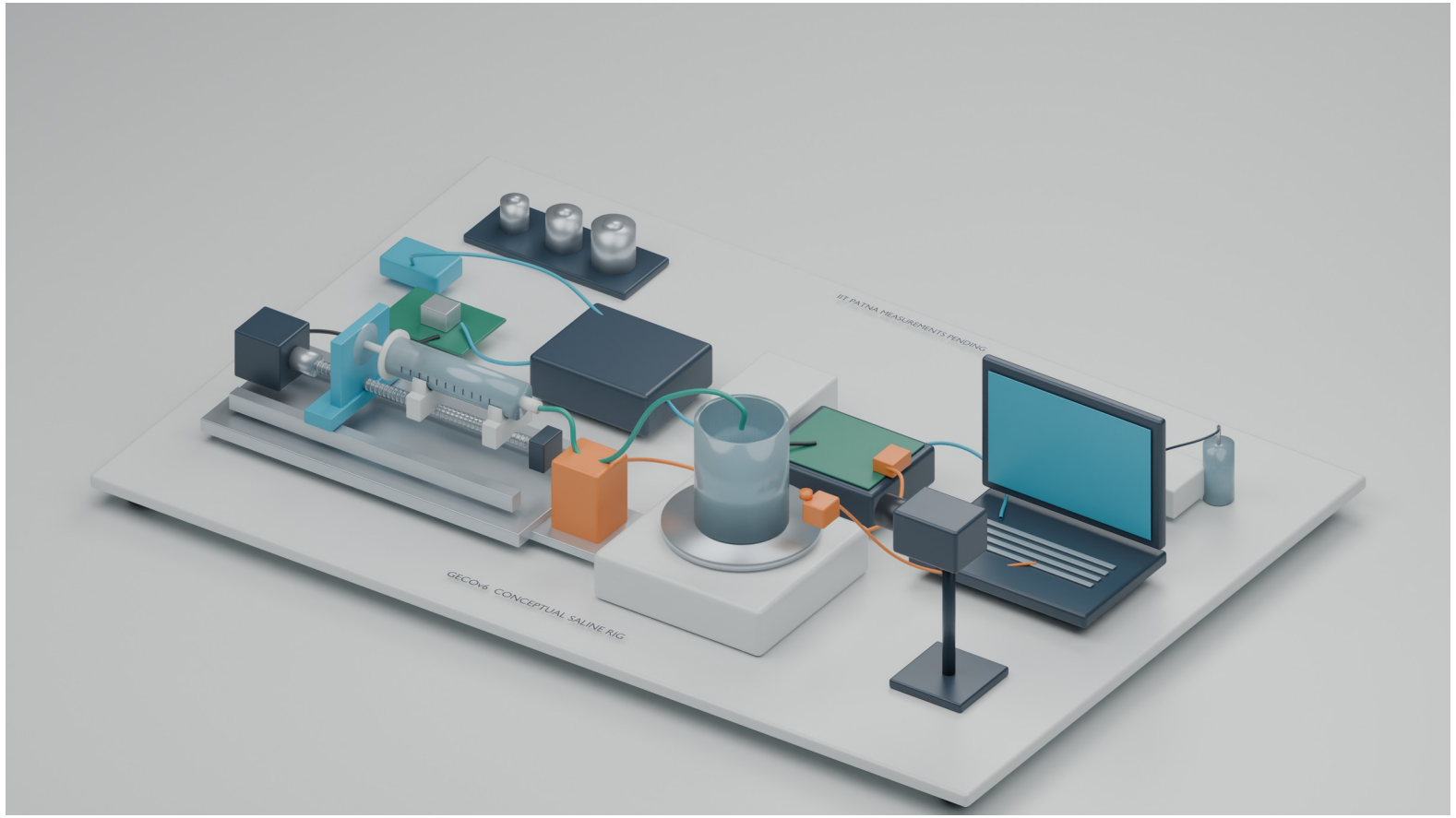
Concept geometry for explanation. Every physical performance estimate is pending.

IIT Patna measurement pending

The complete rig explanation

RIG AND MEASUREMENT

VIDEO IN POWERPOINT



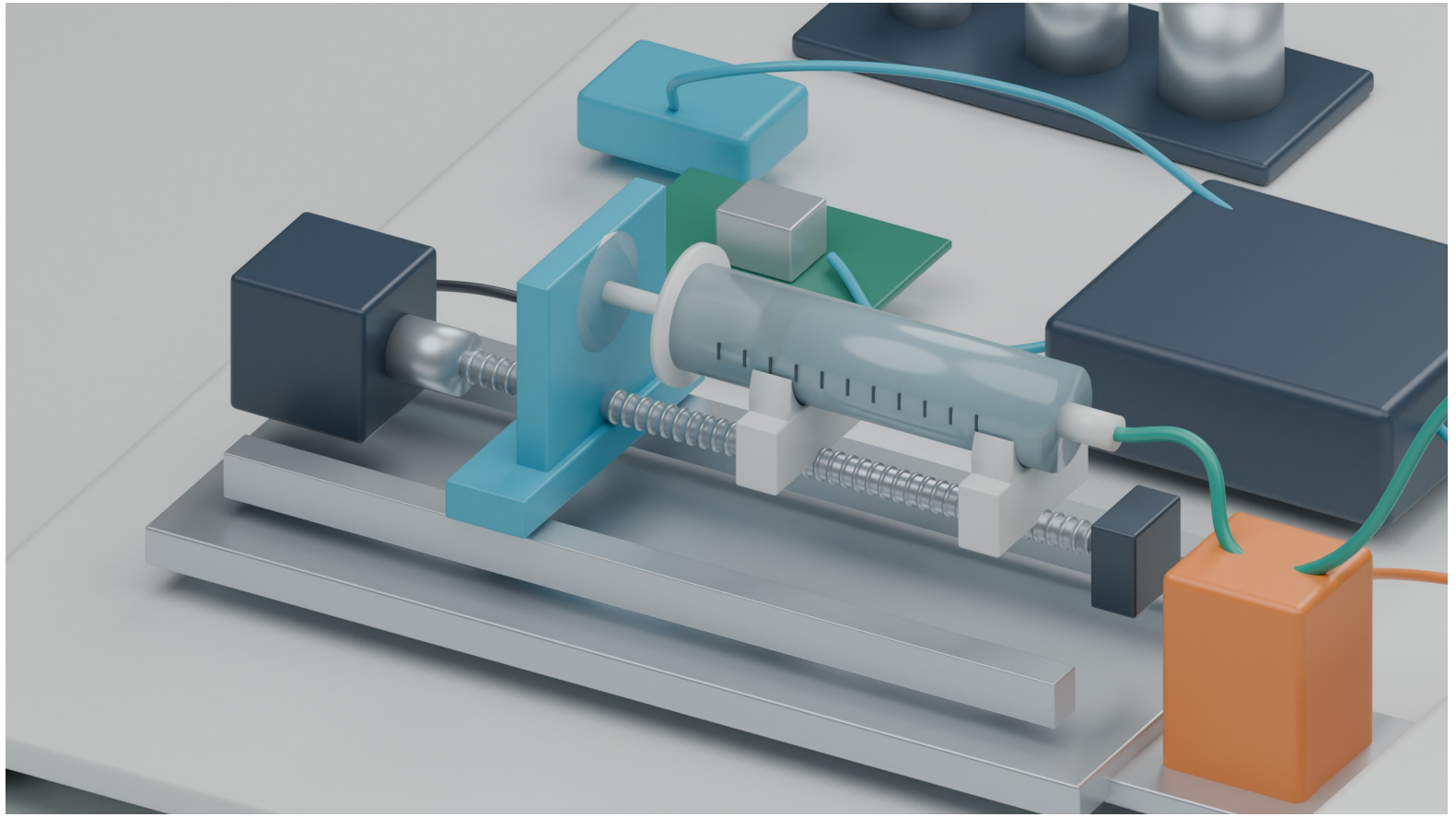
8-minute narrated tour: components, evidence paths, objective measurement and recovery.

IIT Patna measurement pending

Drive mechanism

RIG AND MEASUREMENT

VIDEO IN POWERPOINT



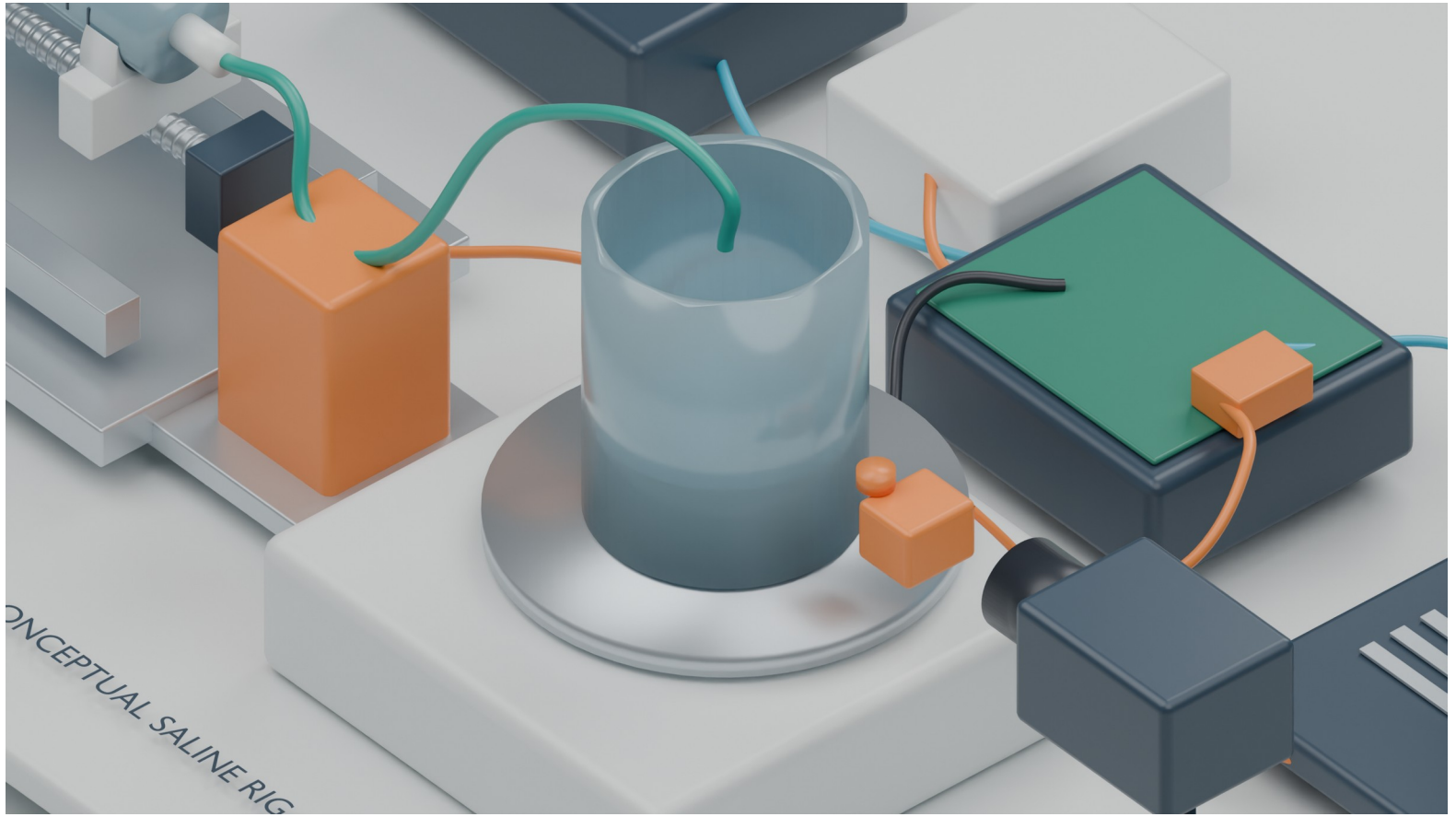
Motor, coupling, lead screw, carriage and rail: verify actual travel and repeatability.

IIT Patna measurement pending

Mass measurement

RIG AND MEASUREMENT

VIDEO IN POWERPOINT



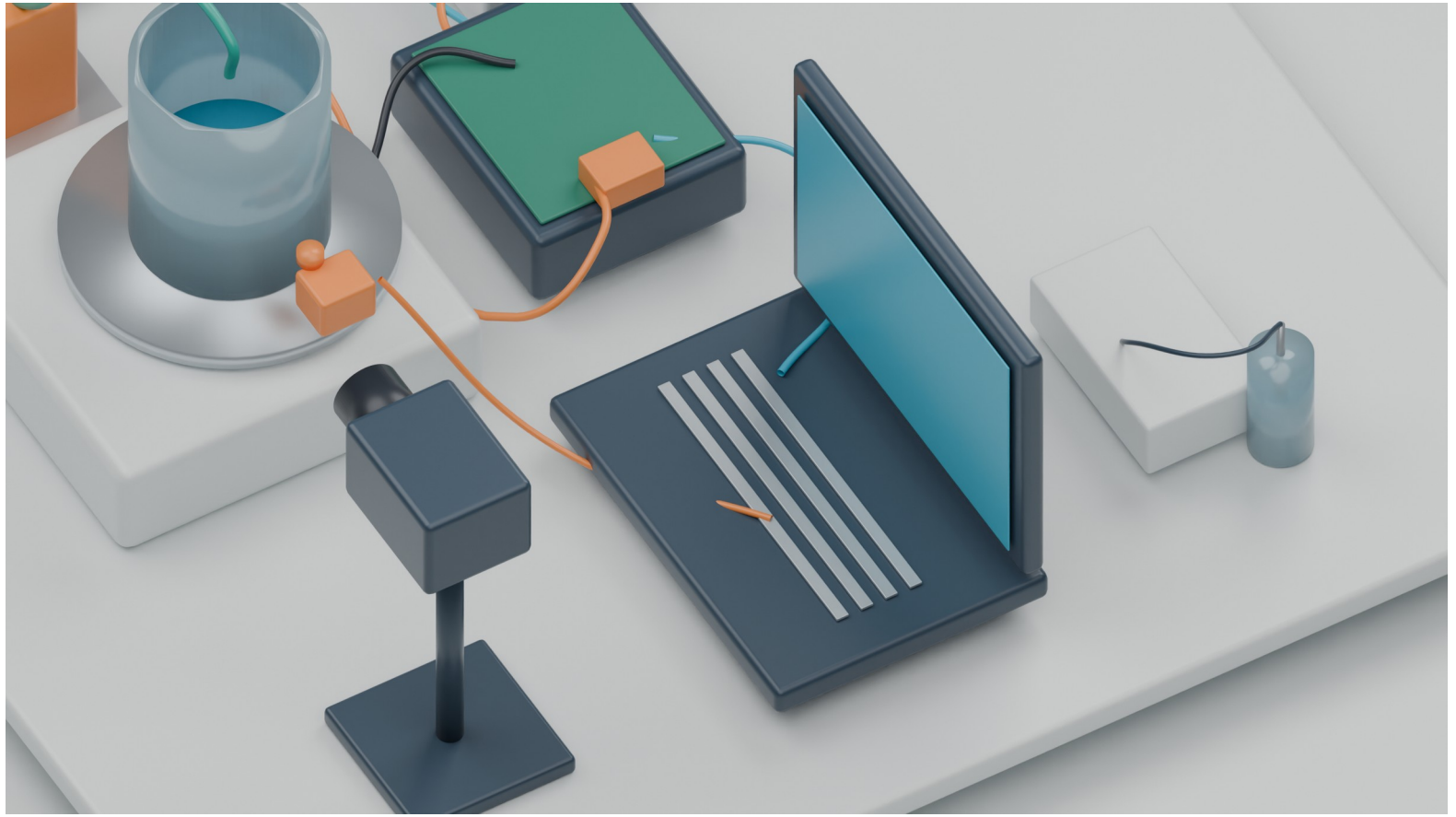
Receiver, load cell and independent reference: define the outlet and retain signed readings.

IIT Patna measurement pending

Independent recording and clocks

RIG AND MEASUREMENT

VIDEO IN POWERPOINT



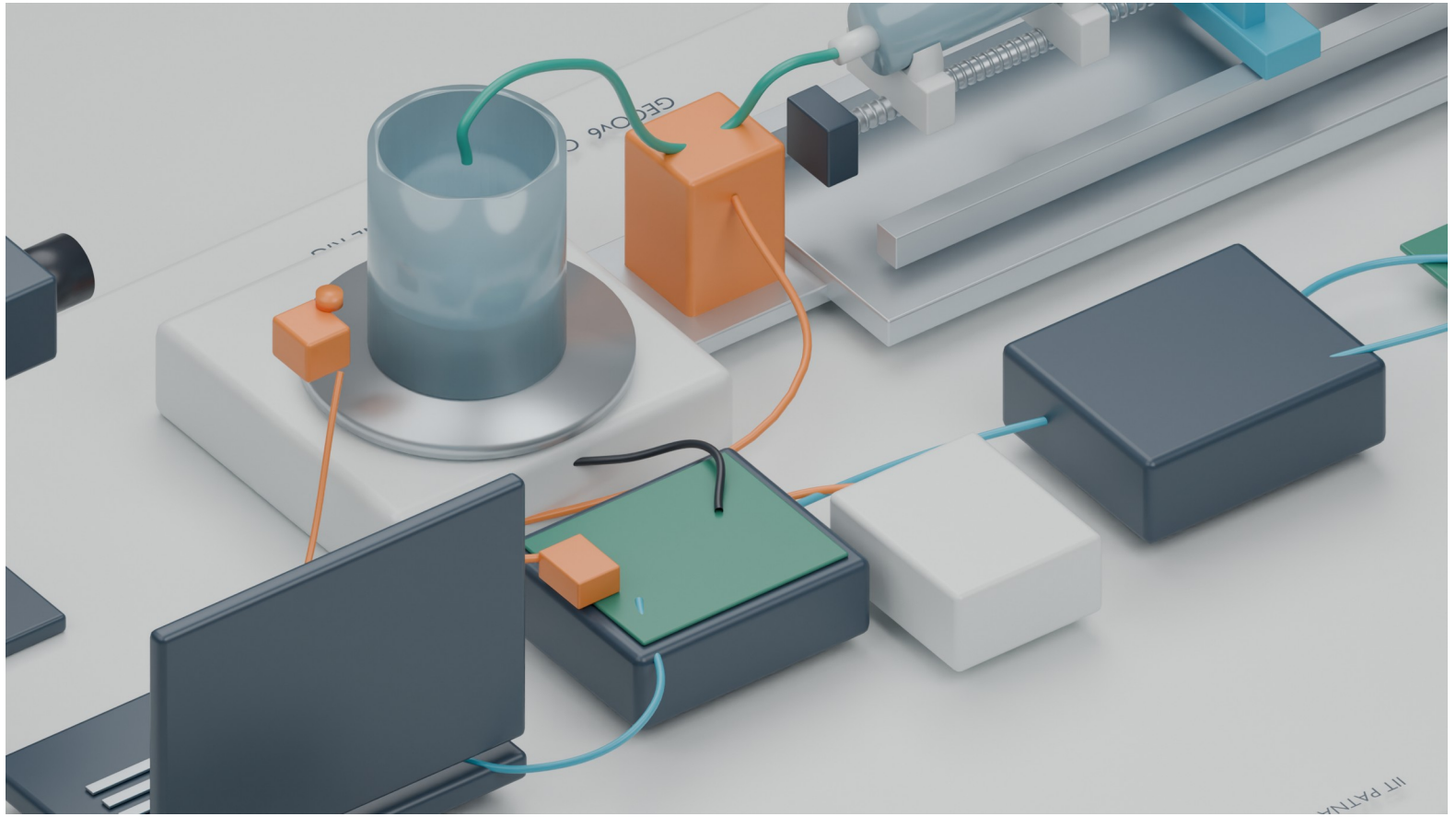
Acquisition computer, camera and timing marker: align clocks with measured residual error.

IIT Patna measurement pending

Local interruption

RIG AND MEASUREMENT

VIDEO IN POWERPOINT

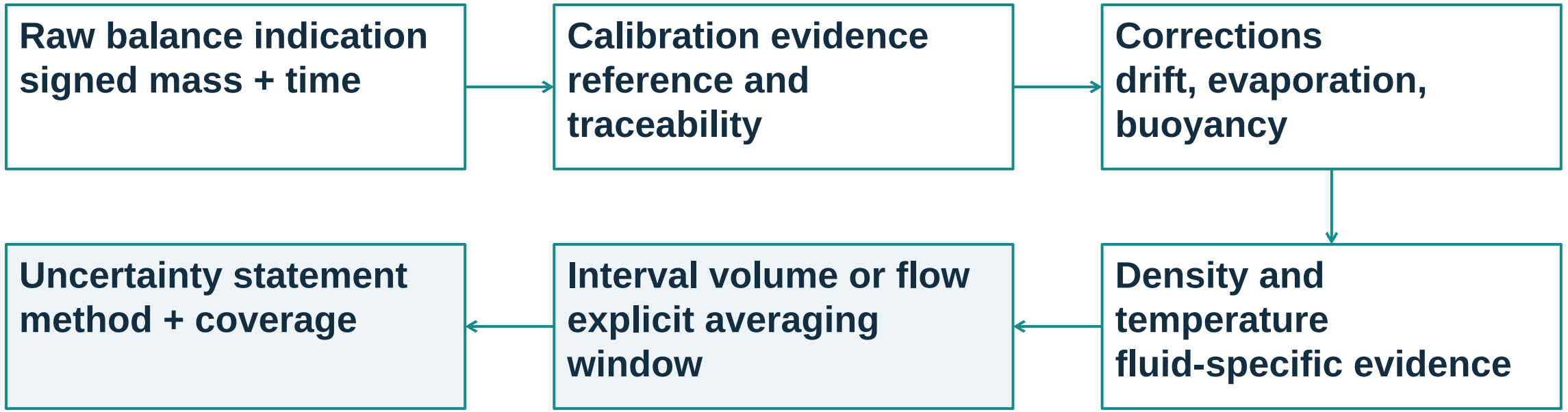


Trip command, actuator disable and outlet cessation are separate observable events.

IIT Patna measurement pending

From indication to a measurement

RIG AND MEASUREMENT



Never convert every negative mass change to zero. It biases low-flow estimates.

IIT Patna measurement pending

Objective measurements at IIT Patna

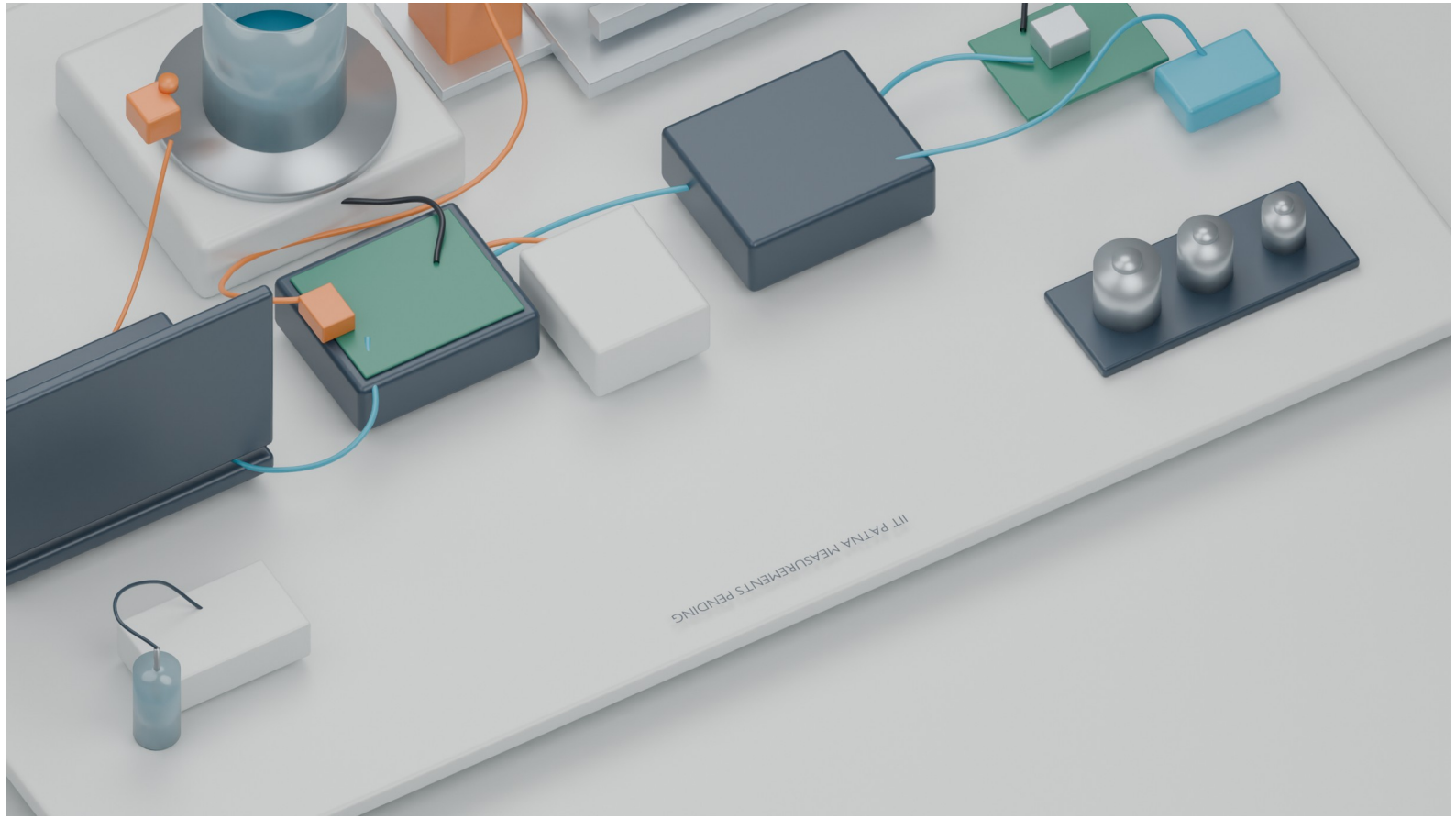
RIG AND MEASUREMENT

Quantity	Independent evidence	Report with estimate
Cumulative delivery	Calibrated gravimetric reference	Bias, repeatability, uncertainty
Low-flow response	Long interval capture; drift/evaporation runs	Window length and detection limit
Trip-to-disable latency	Aligned command and electrical event logs	Clock offset, residual error, interval
Outlet cessation / residual fluid	Defined outlet trace + video if validated	Cessation criterion, tail mass, uncertainty
Recovery	Probe, emission and state records	Deficit, failed restart, confirmation time

IIT Patna measurement pending

Calibration and temperature

RIG AND MEASUREMENT

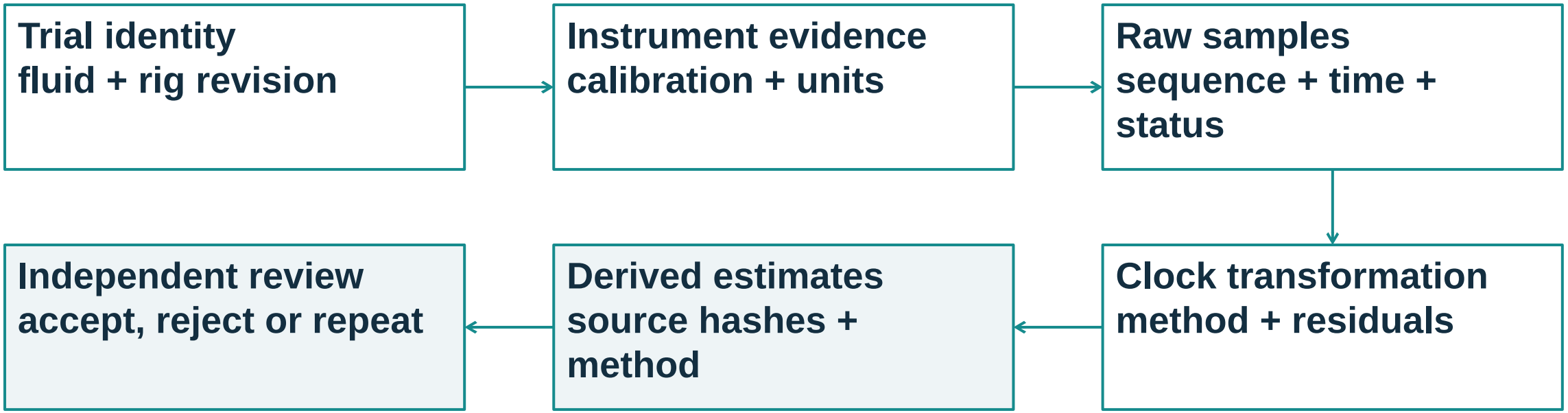


Reference masses and a separate fluid-temperature sample illustrate required evidence.

IIT Patna measurement pending

An immutable acquisition record

IIT PATNA HANDOFF



Blank templates remain blank until measured. The offline validator rejects missing evidence.

IIT Patna measurement pending

Paper fields to fill after the build

IIT PATNA HANDOFF

Physical result section	IIT Patna must supply
Rig and instrumentation	As-built dimensions, parts, fluid, calibration evidence
Measurement performance	Mass/volume bias, repeatability, low-flow behavior
Timing and stopping	Clock alignment, disable latency, outlet cessation, residual mass
Failure and recovery	Excess, deficit, long interruption, failed and successful restart
Transfer and reproducibility	Measured-data replay, model discrepancy, raw traces and analysis

Amber text and tables in the manuscript mark these pending claims beside the relevant discussion.

IIT Patna measurement pending

M01–M08: apparatus and metrology

IIT PATNA HANDOFF

ID	Evidence to complete
M01–M02	As-built apparatus; outlet and acquisition boundary
M03–M04	Traceable balance calibration; fluid temperature and density
M05–M06	Evaporation/environment; mechanical force isolation
M07–M08	Independent reference; reconstruction and zero-flow bias

IIT Patna measurement pending

M09–M16: interruption and transfer

IIT PATNA HANDOFF

ID	Evidence to complete
M09–M10	Delivery accuracy and timing; acquisition synchronization
M11–M12	Stop response; post-stop residual flow
M13–M14	Stop-path independence; nuisance alarms and misses
M15–M16	Restart and recovery; locked physical study and records

IIT Patna measurement pending

What can be claimed now

CONCLUSION

A reproducible simulation and methods contribution with explicit evidence boundaries.

A corrected prediction comparison that favors simpler candidates on these records.

A documented intervention tradeoff and a physical validation protocol ready for the team.

What remains before submission

CONCLUSION

Authors must approve the claims, contributions, references and assistance disclosure.

Choose the IEEE venue, then apply its exact scope, length and submission requirements.

Physical measurement claims remain amber until IIT Patna supplies reviewed evidence.

The research story

CONCLUSION

We asked whether a monitor can connect permission with credible delivery evidence.

We repaired the testable implementation and challenged it against simpler alternatives.

The strongest result is a transparent method, a bounded claim and a concrete next experiment.

Technical: prediction and calibration

TECHNICAL APPENDIX

Item	Definition or limit
Innovation normalization	Inflation scales alarm normalization, not nominal measurement covariance
Forecast candidates	Mechanistic, persistence, bounded causal trend
Gap handling	Filter memory reset after prolonged missing observations
Coverage	Empirical symmetric ±1.96-scale coverage; participant range
Optimization	Bounded fit; near-bound solutions are not identifiability evidence

Technical: model assumptions

TECHNICAL APPENDIX

Item	Frozen illustrative value
Sampling and horizon	5 min; 576 intervals over 48 h
Dose-to-mass mapping	0.1005 g per research U
Actuator	200 steps/U; gain 1.015; positive-command dribble 0.0015 g
Balance	0.01 g quantization; median of 5; 0.003 g sample noise
Mass residual scale	0.0051435068 g; supplied numerical calibration

Technical: uncertainty and pairing

TECHNICAL APPENDIX

Analysis	Unit and meaning
Forecast contrast	Participant paired difference; 10,000 bootstrap resamples
Intervention contrast	Parameter-vector mean over paired seeds; vector bootstrap
Clean comparison	Shared request tape and clean reference emission
Recovery censoring	Reported separately; failed recovery is not a finite time
Multiplicity	Descriptive contrasts; no universal superiority conclusion

Technical: prior work and distinction

TECHNICAL APPENDIX

Source family	What it contributes	GECOv6 boundary
GlucOS; KnowSafe	Safety monitoring and runtime decisions	No first-monitor claim
Hovorka 2004; prediction literature	Physiology and forecasting context	Not clinical model validation
Gravimetric pump metrology	Measurement and uncertainty methods	No transferred instrument accuracy
IEEE author guidance	Format and assistance disclosure	No acceptance or affiliation claim

Technical: originality and provenance

TECHNICAL APPENDIX

Scientific review precedes writing and similarity diagnostics.

The bounded local overlap check is evidence about the supplied corpus only.

AI assistance and synthetic narration are disclosed; no AI-free, SynthID-free or detector certification is claimed.

Technical: reproduce and inspect

TECHNICAL APPENDIX

Frozen inputs, configurations, raw traces and review receipts travel with this package.

The original release is preserved separately; new results are clearly versioned.

Start with the release guide, then follow the exact experiment and artifact instructions.